

Motivating the demonstration of potential project pipelines for OpenSAFELY users

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Background

- Current OpenSAFELY teaching materials cover setting up a dataset definition file using ehrQL (the language created to query data in OpenSAFELY)
- However, project pipelines (the workflow used by researchers for an entire project in OpenSAFELY) vary in complexity and currently examples for entire pipelines exist only in the form of repositories created by researchers for their projects
- In this project we will create three project pipelines which increase in complexity, in order to illustrate the ways in which OpenSAFELY can be used
- To create these pipelines, we will investigate a genuine research question and run the pipelines in the secure environment
- This project will investigate the prescribing behaviour for Salbutamol inhalers surrounding the pandemic
- This is motivated by guidance which recommends patients prescribed a high quantity of inhalers, within 12 months, be flagged for review
 - Currently quantity data is not available for broader research use in OpenSAFELY
 - As a proxy, we have used prescription quantity - within research, while this has been done before, we acknowledge that it may not be a suitable proxy; however, for the purposes of this resource we have made such a decision
- Different NHS formularies have different thresholds for this flagging, so we will investigate the average number of inhalers prescribed within 12 months, surrounding the pandemic

Objectives

1. Create a series of project pipelines to illustrate ways in which OpenSAFELY can be utilised
 - a. *Pipeline 1*: using one time period, and the entire population; containing no parameterised actions
 - b. *Pipeline 2*: using three time periods, and the entire population; containing parameterised ehrQL actions for different time periods
 - c. *Pipeline 3*: using three time periods, with two distinct populations; containing parameterised ehrQL actions for different time periods and parameterised R scripts to run entire analyses separately by age
2. Provide examples of good practice throughout the repository, which will contain all three pipelines, to provide good reference points for researchers
3. Investigate Salbutamol inhaler prescribing surrounding the pandemic
 - a. Initially we will look at inhaler quantity in the general population during the pandemic - *pipeline 1*
 - b. Subsequently we will compare this to how the quantity has changed from before the pandemic, and inhaler quantity post-pandemic - *pipeline 2*
 - c. Finally we will extend this comparison to determine whether changes seen during the pandemic differ amongst younger individuals (more likely to be asthmatic) and older individuals (at higher risk of COPD) - *pipeline 3*

Methods

Data Source

Primary care records managed by the GP software provider, TPP were linked to ONS death data through OpenSAFELY, a data analytics platform created by our team on behalf of NHS England to address urgent COVID-19 research questions (<https://opensafely.org>).

OpenSAFELY provides a secure software interface allowing the analysis of pseudonymized primary care patient records from England in near real-time within the EHR vendor's highly secure data centre, avoiding the need for large volumes of potentially disclosive pseudonymized patient data to be transferred off-site. This, in addition to other technical and organisational controls, minimises any risk of re-identification. Similarly pseudonymized datasets from other data providers are securely provided to the EHR vendor and linked to the primary care data. The dataset analysed within OpenSAFELY is based on 24 million people currently registered with GP surgeries using TPP SystmOne software. It includes pseudonymized data such as coded diagnoses, medications and physiological parameters. No free text data is included. Further details on our information governance can be found [here](#).

Outputs are accessed by researchers as aggregated data with suppression of small numbers and rounding of counts in compliance with re-identification minimisation requirements for OpenSAFELY.

Information Governance

NHS England is the data controller; TPP is the data processor; and the key researchers on OpenSAFELY are acting on behalf of NHS England. This implementation of OpenSAFELY is hosted within the TPP environment, which is accredited to the ISO 27001 information security standard and is NHS IG Toolkit compliant (1,2); patient data have been pseudonymized for analysis and linkage using industry standard cryptographic hashing techniques; all pseudonymized datasets transmitted for linkage onto OpenSAFELY are encrypted; access to the platform is through a virtual private network (VPN) connection; the researchers hold contracts with NHS England and only access the platform to initiate database queries and statistical models; all database activity is logged; and only aggregate statistical outputs leave the platform environment following best practice for anonymization of results such as statistical disclosure control for low cell counts (3). The OpenSAFELY research platform adheres to the data protection principles of the UK Data Protection Act 2018 and the EU General Data Protection Regulation (GDPR) 2016. In March 2020, the Secretary of State for Health and Social Care used powers under the UK Health Service (Control of Patient Information) Regulations 2002 (COPI) to require organisations to process confidential patient information for the purposes of protecting public health, providing healthcare services to the public and monitoring and managing the COVID-19 outbreak and incidents of exposure (4). Together, these provide the legal bases to link patient datasets on the OpenSAFELY platform. GP practices, from which the primary care data are obtained, are

required to share relevant health information to support the public health response to the pandemic, and have been informed of the OpenSAFELY analytics platform.

Study Design and Population

Study Period

Pipeline 1

- March 2020 - Feb 2022
- Participants will be 'recruited' at the beginning of March 2020. Individuals will be followed up until the date of death, practice deregistration, or end of the study period (28 February 2022).

Pipelines 2 and 3

- March 2017 - Feb 2025
 - Pre-pandemic: March 2017 - Feb 2020
 - Pandemic: March 2020 - Feb 2022
 - Post-pandemic: March 2022 - Feb 2025
- 'Recruitment' will be performed at the beginning of each period (1 March 2017, 2020, and 2022 respectively), participants will be included in multiple periods as long as they meet the inclusion criteria for each season. Individuals will be followed up until the date of death, practice deregistration, or end of the study period (29 February 2020, 28 February 2022, and 28 February 2025 respectively).

Inclusion criteria

Pipelines 1 and 2

- Participants will be included if they have at least three months of continuous practice registration prior to the start of the study period
- Participants will be limited to those aged 12 and over due to the differences in asthma management in children under 12 (5)

Pipeline 3

- Participants will be included if they have at least three months of continuous practice registration prior to the start of the study period
- Two cohorts will be analysed in pipeline 3, chosen from research into the age distribution of patients with COPD (6,7):
 - Younger individuals: aged 12-59
 - Older individuals: aged 60+

Exclusion Criteria

We will exclude individuals with missing data on age and sex, due to the likely poor data quality of these records.

Study Measures

Patient Characteristics

The following patient characteristics will be included in analyses:

- Age: 12-100
- Sex: female, male, intersex
- Diagnosis of respiratory condition:
 - Asthma
 - COPD
- Other
 - IMD
 - Ethnicity

Inhaler Quantity

We will investigate the number of salbutamol inhalers prescribed per patient annually, using prescriptions as a proxy for quantity - with the awareness that multiple inhalers could be included in one prescription.

Identifying Patients and Inhalers

We will use existing codelists from OpenCodelists to identify both patient characteristics and inhaler prescriptions:

Identifying factor	Codelist
Asthma diagnosis	Asthma diagnosis codes , Asthma medications (inhalers) , Asthma medications (oral steroids)
COPD diagnosis	COPD codelist , COPD medications , COPD resolved codes
Salbutamol	https://www.opencodelists.org/codelist/nhs-drug-refsets/gensalbmddinhdrug_cod/20250627/ https://www.opencodelists.org/codelist/nhs-drug-refsets/gensalbdpinhdrug_cod/20250627/ https://www.opencodelists.org/codelist/nhs-drug-refsets/ventevobrdinhdrug_cod/20250627/ https://www.opencodelists.org/codelist/nhs-drug-refsets/ventaccubrdinhdrug_cod/20250627/ https://opencodelists.org/codelist/nhs-drug-refsets/salbbdrinhdrug_cod/20250627/ https://www.opencodelists.org/codelist/nhs-drug-refsets/salabrdinhdrug_cod/20250627/ https://www.opencodelists.org/codelist/nhs-drug-refsets/salabrdinhdrug_cod/20250627/

	g-refsets/easybrdinhdru_cod/20250627/
	https://www.opencodelists.org/codelist/nhs-drug-refsets/airobrdinhdru_cod/20250627/
	https://www.opencodelists.org/codelist/nhs-drug-refsets/airsbrdinhdru_cod/20250627/

Table 1: list of existing codelists available for use in this project and their corresponding link

To ensure codelists will be suitable for the entire study period we will be using NHS refsets, the newest version of these in SNOMED CT will contain inactive codes which may exist in older records, for medications codes are retrospectively updated in TPP records and hence the newest codelists should be used.

Statistical Analysis

Exploratory Analyses

Initially the general population will be explored to look at the characteristics of those requiring inhalers, e.g.

- Proportion with an existing diagnosis of either asthma or COPD
- Average age of those needing inhalers vs general population

The population will then be reduced to those with a prescription of salbutamol in the two years preceding the study start, and some descriptive work will be performed to look at inhaler quantity distribution by:

- Age: grouped
- Sex: female, male, intersex
- Condition: asthma or COPD

In *pipeline 2* these descriptions will be separate for the different study periods. In *pipeline 3* these descriptions will be separate for different age cohorts and study periods.

Primary Analyses

Regression models will be run to explore the relationship between inhaler quantity and the following:

- Age: continuous
- Sex: factor
- Presence of respiratory condition diagnosis:
 - Asthma: factor
 - COPD: factor
- Other:
 - IMD: factor

- Ethnicity: factor

In *pipeline 2* these analyses will be performed separately in the different study periods. In *pipeline 3* these analyses will be performed separately across age cohorts and study periods.

Software and Reproducibility

Data management will be performed using Python, with analysis carried out using Python/R.

References

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3. NHS Digital [Internet]. [cited 2023 Oct 16]. ISB1523: Anonymisation Standard for Publishing Health and Social Care Data. Available from: <https://digital.nhs.uk/data-and-information/information-standards/information-standards-and-data-collections-including-extractions/publications-and-notifications/standards-and-collections/isb1523-anonymisation-standard-for-publishing-health-and-social-care-data>
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5. Recommendations | Asthma: diagnosis, monitoring and chronic asthma management (BTS, NICE, SIGN) | Guidance | NICE [Internet]. NICE; 2024 [cited 2025 Oct 3]. Available from: <https://www.nice.org.uk/guidance/ng245/chapter/recommendations#pharmacological-management-in-people-aged-12-and-over>
6. Stone PW, Osen M, Ellis A, Coaker R, Quint JK. Prevalence of Chronic Obstructive Pulmonary Disease in England from 2000 to 2019. *Int J Chron Obstruct Pulmon Dis*. 2023 July 21;18:1565–74.
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